

Companion Growth & Graduation

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Two behavioral pillars from the original concept, grounded entirely in real stock activity — no blockchain, no tokens. Now implemented in the MVP.

What We Built

The companion and the Graduation / Hall of Fame ideas, refined and wired to real portfolio behavior. A companion's state is a live reflection of its holding's actual story — how long it has been held, how much has been added, dividends captured, and drawdowns weathered. Selling is reframed as a dignified **graduation** that records the position's real realized performance.

Design Principle

Every growth signal and reward derives from **real holding data** (holding days, DCA purchases, dividends, drawdown, realized return) — never from in-app tokens or speculation. Healthy investing habits are rewarded; blind attachment is checked by an honest health warning.

1 Pillar 1 — Companion Growth → Stock Behavior

The companion evolves on a predictable time-based cycle (Day 0 / 30 / 90), while concrete investing actions add living detail on top:

Growth Signal	Real Stock Data	In-App Expression	Status
Holding duration	Days since first buy (frozen at sale)	Base evolution stage: Seed → Sprout → Sapling	✓ Built
Nutritional Feed (DCA)	Additional purchases (date, shares, cost)	Logged catalyst; recomputes avg cost & total shares	✓ Built
Dividends	Dividend payouts (date, amount)	'Golden Berries' captured; counts toward total return	✓ Built
Weathering volatility	Held through a drawdown of 15% or more	'Diamond Hands' resilience badge (persisted)	✓ Built
Fundamental concern	Current drawdown of 30% or more	'Needs a checkup' health warning — reassess, don't cling	✓ Built

Recording a position's current market value unlocks a live Performance panel: unrealized return (including dividends), annualized CAGR, and drawdown.

2 Pillar 2 — Graduation & Hall of Fame

Selling never kills a companion — it **graduates**. The ceremony branches on the real outcome:

Ascension Ceremony

A non-negative total return (and not a stop-loss). The companion ascends to the Hall of Fame as a permanent trophy recording its real metrics.

Peaceful Relocation

A stop-loss, or a sale at a net loss. A dignified send-off that archives the lesson without guilt — disciplined selling is honored too.

Hall of Fame Trophy — Recorded Metrics

Metric	Definition (from real data)
Total Return	<code>soldPrice + dividends - total invested (with DCA)</code> , as \$ and %
Annualized (CAGR)	Compound annual growth rate over the holding period
Dividends Captured	Sum of all dividend payouts recorded while holding
Held	First buy date → sale date (frozen)
Exit Type	Ascension or Peaceful Relocation

3 Formulas & Rules

Quantity	Rule
Total Invested	<code>costBasis + sum(additionalBuys.cost)</code>
Total Return	<code>(soldPrice or currentValue) + dividends - totalInvested</code>
Return %	<code>totalReturn / totalInvested</code>
CAGR	<code>(finalValue / totalInvested) ^ (365 / days) - 1</code>
Drawdown	<code>(currentValue - totalInvested) / totalInvested</code>
Diamond Hands	<code>weathered drawdown ≤ -15% (persisted)</code>
Health Warning	<code>live drawdown ≤ -30%</code>
Exit Type	<code>stop-loss → relocation; else totalReturn ≥ 0 → ascension</code>

4 Data Model Additions

New optional fields on the `Holding` type — fully backward-compatible with existing saved portfolios:

Field	Type	Purpose
<code>additionalBuys?</code>	<code>BuyLot[]</code>	DCA purchases after the first buy
<code>dividends?</code>	<code>DividendRecord[]</code>	Dividend payouts captured while holding
<code>currentValue?</code>	<code>number</code>	Manually-recorded current market value
<code>weatheredDip?</code>	<code>boolean</code>	Set once a 15%+ drawdown is held through
<code>sellReason?</code>	<code>SellReason</code>	profit-target stop-loss rebalance

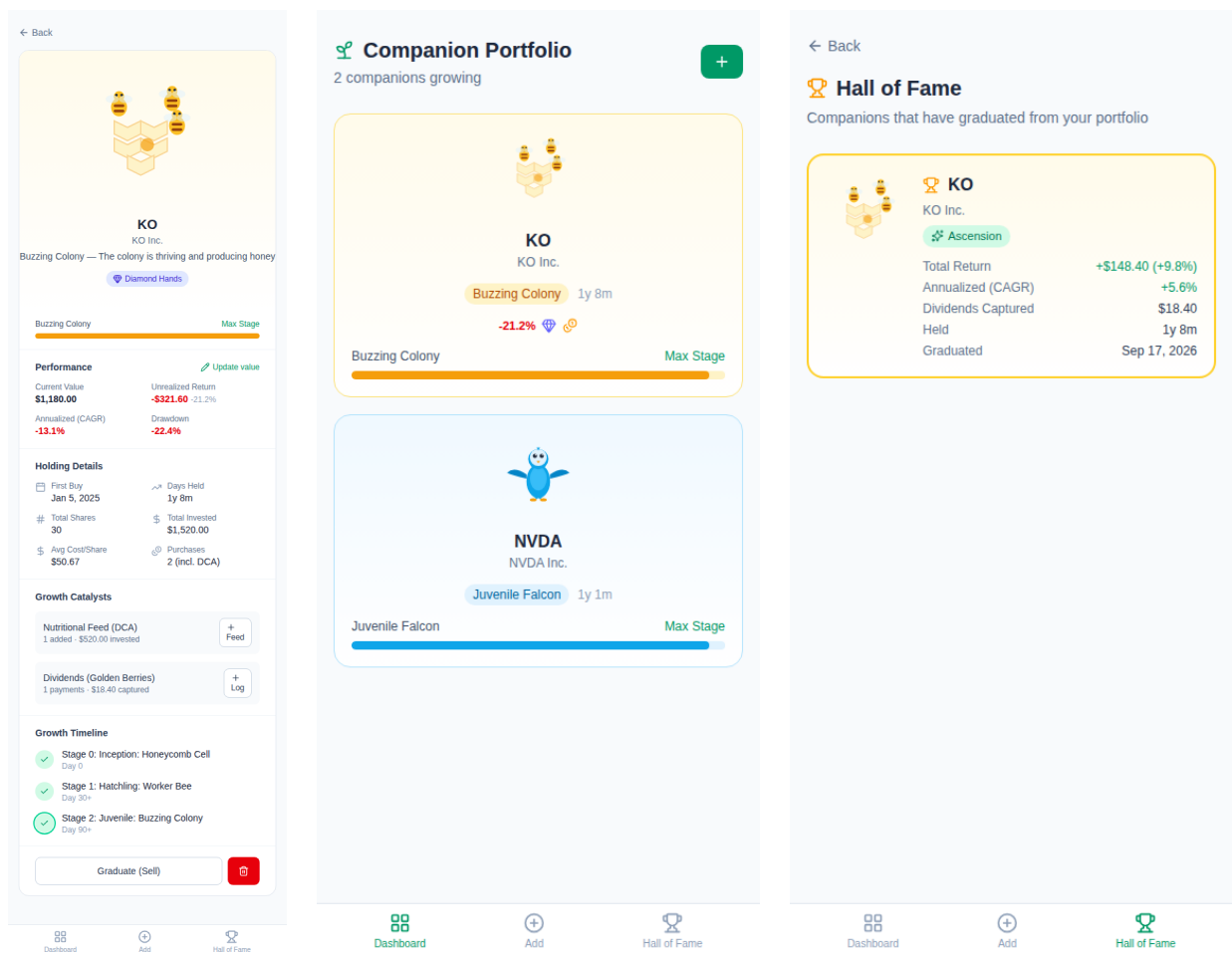
5 Implementation Map

Area	Files
Types & store	<code>types/holding.ts</code> , <code>store/holdingStore.ts</code> (<code>addBuy</code> , <code>addDividend</code> , <code>updateCurrentValue</code> , <code>sellHolding+reason</code>)
Performance logic	<code>utils/performance.ts</code> (<code>returns</code> , <code>CAGR</code> , <code>drawdown</code> , <code>health</code> , <code>exit type</code>)

Area	Files
Growth hook	hooks/useCompanion.ts (holding days frozen at sale)
Logging UI	forms/AddBuyModal.tsx, forms/AddDividendModal.tsx, forms/SellHoldingModal.tsx
Screens	pages/HoldingDetail.tsx, cards/HallOfFameCard.tsx, cards/CompanionCard.tsx

Verified: TypeScript build passes, linter clean, and every flow above was exercised end-to-end in a real browser (screenshots follow).

6 The Built Feature



Holding detail: performance, catalysts, badges

Dashboard: live return & status badges

Hall of Fame: Ascension trophy with real metrics

Example: KO held ~1y 8m, one DCA purchase, one dividend, weathered a dip (Diamond Hands), then graduated via Ascension at a +\$148.40 total return (+9.8%, +5.6% CAGR, \$18.40 dividends captured).